

Complete Ollama + Open WebUI Installation Guide

Ubuntu Server 22.04 / 24.04 (Manual Installation from Downloaded File)

This guide assumes:

- Ubuntu Server is freshly installed
- Docker is installed
- You already downloaded:

```
ollama-linux-amd64.tar.zst
```

- Server IP example:

```
172.17.90.207
```

- Goal:
 - Install Ollama
 - Run AI models locally
 - Install Open WebUI
 - Access from browser
 - Prepare for Nextcloud AI integration

Based on Ollama's Linux documentation and manual installation process.

Step 1 - Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
sudo reboot
```

Reconnect after reboot.

Step 2 - Install Required Packages

```
sudo apt update
sudo apt install -y curl wget zstd unzip tar
```

Verify:

```
zstd --version
```

Step 3 - Go To Download Directory

Example:

```
cd ~/ollama
```

Verify:

```
ls -lh
```

You should see:

```
ollama-linux-amd64.tar.zst
```

Step 4 - Extract Ollama

Extract into system directories:

```
sudo tar --use-compress-program=unzstd \  
-xvf ollama-linux-amd64.tar.zst \  
-C /usr/local
```

Verify:

```
ls -l /usr/local/bin/ollama
```

Expected:

```
/usr/local/bin/ollama
```

Step 5 - Verify Installation

```
ollama --version
```

Example:

```
ollama version 0.x.x
```

Step 6 - Create Ollama Service Account

Official Ollama documentation recommends running as a dedicated service account.

```
sudo useradd -r \  
-s /bin/false \  
-U \  
-m \  
-d /usr/share/ollama \  
ollama
```

Verify:

```
id ollama
```

Step 7 - Create Model Storage

```
sudo mkdir -p /home/ollama/.ollama  
sudo chown -R ollama:ollama /home/ollama/.ollama
```

Step 8 - Create Systemd Service

Create:

```
sudo nano /etc/systemd/system/ollama.service
```

Paste:

```
[Unit]
Description=Ollama Service
After=network-online.target

[Service]
ExecStart=/usr/local/bin/ollama serve
User=ollama
Group=ollama
Restart=always
RestartSec=3

Environment="PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
Environment="OLLAMA_HOST=0.0.0.0:11434"

[Install]
WantedBy=multi-user.target
```

Save and exit.

Step 9 - Enable Service

```
sudo systemctl daemon-reload
sudo systemctl enable ollama
sudo systemctl start ollama
```

Check status:

```
sudo systemctl status ollama
```

Expected:

```
active (running)
```

Step 10 - Verify API Port

```
sudo ss -lntp | grep 11434
```

Expected:

```
*:11434
```

Test API:

```
curl http://127.0.0.1:11434/api/tags
```

Expected:

```
{"models": []}
```

Step 11 - Install First AI Model

For your 16GB RAM server I recommend:

Qwen 3 4B

```
ollama pull qwen3:4b
```

Verify:

```
ollama list
```

Expected:

```
NAME
qwen3:4b
```

Step 12 - Test Model

```
ollama run qwen3:4b
```

Type:

```
hello
```

Expected:

```
Hello! How can I help you?
```

Exit:

```
/bye
```

Step 13 - Install Docker

If Docker is not installed:

```
curl -fsSL https://get.docker.com | sh
```

Verify:

```
docker --version
```

Step 14 - Install Open WebUI

Create persistent storage:

```
docker volume create open-webui
```

Run container:

```
docker run -d \
--name open-webui \
--restart unless-stopped \
-p 3000:8080 \
-v open-webui:/app/backend/data \
ghcr.io/open-webui/open-webui:main
```

Check:

```
docker ps
```

Expected:

```
open-webui
Up
0.0.0.0:3000->8080
```

Step 15 - Open Browser

Navigate to:

```
http://SERVER-IP:3000
```

Example:

```
http://172.17.90.207:3000
```

Create:

- Administrator Account
 - Username
 - Password
-

Step 16 - Connect Open WebUI To Ollama

Go to:

```
Admin Panel
→ Settings
→ Connections
→ Ollama
```

Enter:

```
http://SERVER-IP:11434
```

Example:

`http://172.17.90.207:11434`

DO NOT use:

`http://127.0.0.1:11434`

because Open WebUI runs inside Docker.

Step 17 - Test AI

Create new chat.

Choose:

`qwen3:4b`

Send:

`Hello`

You should receive a response in a few seconds.

Useful Commands

List Models

```
ollama list
```

Remove Model

```
ollama rm qwen3:4b
```

Show Running Models

```
ollama ps
```

View Logs

```
sudo journalctl -u ollama -f
```

Restart Ollama

```
sudo systemctl restart ollama
```

Restart Open WebUI

```
docker restart open-webui
```

Recommended Models For KAYSON

Use Case	Model
General Chat	qwen3:4b
Better Quality	qwen3:8b
Coding	qwen2.5-coder:7b
Persian + English	qwen3:4b
Company Knowledge Base	qwen3:8b
RAG / Documents	qwen3:8b

With your current **16 GB RAM server**, start with:

```
ollama pull qwen3:4b
```

and later move to:

```
ollama pull qwen3:8b
```